TP5 : Environnements de lab reproductibles avec Vagrant

1. vérifier que les extensions de virtualisation du CPU

```
arij@vagrant:~$ grep -Ec '(vmx|svm)' /proc/cpuinfo
8
arij@vagrant:~$
```

Partie_0 : Installation de VirtualBox, Vagrant et Ansible :

```
arij@vagrant:~$ sudo apt install -y virtualbox ansible curl gpg
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
curl est déjà la version la plus récente (8.5.0-2ubuntu10.16).
curl passé en « installé manuellement ».
gpg est déjà la version la plus récente (2.4.4-2ubuntu17.6).
gpg passé en « installé manuellement ».
Les paquets supplémentaires suivants seront installés :
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf ansible-core at-spi2-common at-spi2-core binutils
  binutils-common binutils-x86-64-linux-gnu build-essential bzip2 cpp cpp-13 cpp-13-x86-64-linux-gnu
  cpp-x86-64-linux-gnu dconf-gsettings-backend dconf-service dkms dpkg-dev fakeroot fontconfig g++ g++-13
  g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu
  gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan8 libasound2-data libasound2t64 libasyncns0
  libatk-bridge2.0-0t64 libatk1.0-0t64 libatomic1 libatspi2.0-0t64 libavahi-client3 libavahi-common-data
  libavahi-common3 libbinutils libcairo-gobject2 libcairo2 libcc1-0 libcolord2 libctf-nobfd0 libctf0 libcups2t64
  libdatrie1 libdconf1 libdecor-0-0 libdecor-0-plugin-1-gtk libdouble-conversion3 libdpkg-perl libdrm-amdgpu1
  libdrm-intel1 libegl-mesa0 libegl1 libepoxy0 libfakeroot libfile-fcntllock-perl libflac12t64 libgbm1 libgcc-13-dev
  libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgl1 libgl1-mesa-dri libglvnd0 libglx-mesa0
  libglx0 libgomp1 libgprofng0 libgraphite2-3 libgsoap-2.8.132t64 libgtk-3-0t64 libgtk-3-bin libgtk-3-common
  libharfbuzz0b libhwasan0 libice6 libinput-bin libinput10 libisl23 libitm1 liblcms2-2 libllvm20 liblsan0 liblzf1
  libmd4c0 libmp3lame0 libmpc3 libmpg123-0t64 libmtdev1t64 libogg0 libopus0 libpango-1.0-0 libpangocairo-1.0-0
  libpangoft2-1.0-0 libpciaccess0 libpcre2-16-0 libpixman-1-0 libpulse0 libqt5core5t64 libqt5dbus5t64 libqt5gui5t64
  libqt5help5 libqt5network5t64 libqt5opengl5t64 libqt5printsupport5t64 libqt5qml5 libqt5qmlmodels5 libqt5quick5
```

```
arij@vagrant:~$ wget -O- https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-archive-keyring
.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashic
orp.com $(lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/hashicorp.list
sudo apt update && sudo apt install -y vagrant
--2026-10-22 15:31:19--  https://apt.releases.hashicorp.com/gpg
Resolving apt.releases.hashicorp.com (apt.releases.hashicorp.com)... 18.172.213.91, 18.172.213.113, 18.172.213.100, ...
Connecting to apt.releases.hashicorp.com (apt.releases.hashicorp.com)|18.172.213.91|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1725 (1,7K) [application/x-www-form-urlencoded]
Saving to: 'STDOUT'

-                                   100%[=========================================================>]   1,68K  --.-KB/s    in 0s

2026-10-22 15:31:19 (252 MB/s) - written to stdout [1725/1725]

deb [arch=amd64 signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com noble main
Atteint :1 http://security.ubuntu.com/ubuntu noble-security InRelease
Réception de :2 https://apt.releases.hashicorp.com noble InRelease [12,8 kB]
Réception de :3 https://apt.releases.hashicorp.com noble/main amd64 Packages [329 kB]
Atteint :4 http://tn.archive.ubuntu.com/ubuntu noble InRelease
Atteint :5 http://tn.archive.ubuntu.com/ubuntu noble-updates InRelease
Atteint :6 http://tn.archive.ubuntu.com/ubuntu noble-backports InRelease
342 ko réceptionnés en 2s (186 ko/s)
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
75 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les voir.
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les NOUVEAUX paquets suivants seront installés :
  vagrant
```

Vérification :

```
arij@vagrant:~$ VBoxManage --version
7.0.16_Ubuntur162802
arij@vagrant:~$ vagrant --version
Vagrant 2.4.9
arij@vagrant:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/arij/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/arij/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Aug 31 2026, 10:18:26) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
arij@vagrant:~$
```

Partie_1 : Démo Vagrant :

1. Créer le dossier et la page web

```
arij@vagrant:~$ mkdir -p ~/vagrant_demo/web && cd ~/vagrant_demo
echo "<h1>Bonjour depuis Vagrant</h1>" > web/index.html
arij@vagrant:~/vagrant_demo$
```

2.Créer Vagrantfile :

```
  GNU nano 7.2                                                  Vagrantfile *
# vi: ft=ruby:ts=2:ai:sw=2
$script = <<-SCRIPT
echo "Provisionnement de l'OS"
sudo apt-get update
sudo apt-get install -y apache2
SCRIPT

Vagrant.configure("2") do |config|
  config.vm.box = "bento/ubuntu-24.04"
  config.vm.provision "shell", inline: $script
  config.vm.synced_folder "web/", "/var/www/html"
  config.vm.network "forwarded_port", guest: 80, host: 8080, host_ip: "0.0.0.0"

  config.vm.hostname = "vagrant-web1"
  config.vm.provider "virtualbox" do |vb|
    vb.name = "vagrant-web1"
    vb.memory = 1024
    vb.cpus = 2
  end
end
```

3. Lancer la machine

```
arij@vagrant:~/vagrant_demo$ rm -rf ~/.vagrant.d/tmp/*
vagrant up
Bringing machine 'default' up with 'virtualbox' provider...
==> default: Box 'bento/ubuntu-24.04' could not be found. Attempting to find and install...
    default: Box Provider: virtualbox
    default: Box Version: >= 0
==> default: Loading metadata for box 'bento/ubuntu-24.04'
    default: URL: https://vagrantcloud.com/api/v2/vagrant/bento/ubuntu-24.04
==> default: Adding box 'bento/ubuntu-24.04' (v202510.26.0) for provider: virtualbox (amd64)
    default: Downloading: https://vagrantcloud.com/bento/boxes/ubuntu-24.04/versions/202510.26.0/providers/virtualbox/amd64/vagrant.b
ox
==> default: Successfully added box 'bento/ubuntu-24.04' (v202510.26.0) for 'virtualbox (amd64)'!
==> default: Importing base box 'bento/ubuntu-24.04'...
==> default: Matching MAC address for NAT networking...
==> default: Checking if box 'bento/ubuntu-24.04' version '202510.26.0' is up to date...
==> default: Setting the name of the VM: vagrant-web1
Vagrant is currently configured to create VirtualBox synced folders with
the `SharedFoldersEnableSymlinksCreate` option enabled. If the Vagrant
guest is not trusted, you may want to disable this option. For more
information on this option, please refer to the VirtualBox manual:

  https://www.virtualbox.org/manual/ch04.html#sharedfolders

This option can be disabled globally with an environment variable:

  VAGRANT_DISABLE_VBOXSYMLINKCREATE=1

or on a per folder basis within the Vagrantfile:

  config.vm.synced_folder '/host/path', '/guest/path', SharedFoldersEnableSymlinksCreate: false
==> default: Clearing any previously set network interfaces...
==> default: Preparing network interfaces based on configuration...
    default: Adapter 1: nat
```

4. Tester le site web

```
arij@vagrant:~/vagrant_demo$ curl http://localhost:8080
<h1>Bonjour depuis Vagrant</h1>
arij@vagrant:~/vagrant_demo$
```



Bonjour depuis Vagrant

5.Déteruire la vm

```
arij@vagrant:~/vagrant_demo$ vagrant destroy -f
==> default: Forcing shutdown of VM...
==> default: Destroying VM and associated drives...
arij@vagrant:~/vagrant_demo$
```

Partie_2 Lab : créer un environnement de lab virtuel :

1. Cloner dépôt TP4 :

```
arij@vagrant:~/vagrant_demo$ mkdir -p ~/vagrant_lab && cd ~/vagrant_lab
arij@vagrant:~/vagrant_lab$ git clone https://github.com/arijbelaid/devops_tp4.git temp_tp4
Cloning into 'temp_tp4'...
remote: Enumerating objects: 67, done.
remote: Counting objects: 100% (67/67), done.
remote: Compressing objects: 100% (49/49), done.
remote: Total 67 (delta 1), reused 63 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (67/67), 2.88 MiB | 4.48 MiB/s, done.
Resolving deltas: 100% (1/1), done.
arij@vagrant:~/vagrant_lab$
```

2. Copie les fichiers nécessaires dans ton dossier vagrant_lab

```
arij@vagrant:~/vagrant_lab$ cp -r temp_tp4/roles .
arij@vagrant:~/vagrant_lab$ cp temp_tp4/Dockerfile temp_tp4/docker-compose.yml temp_tp4/index.html .
arij@vagrant:~/vagrant_lab$
```

3.Vérification

```
arij@vagrant:~/vagrant_lab$ ls -l
total 16
-rw-rw-r-- 1 arij arij   79 oct.  22 16:04 docker-compose.yml
-rw-rw-r-- 1 arij arij   67 oct.  22 16:04 Dockerfile
-rw-rw-r-- 1 arij arij  168 oct.  22 16:04 index.html
drwxrwxr-x 5 arij arij 4096 oct.  22 16:04 roles
arij@vagrant:~/vagrant_lab$
```

4. Créer le playbook et le Vagrantfile

```
 GNU nano 7.2                                                playbook.yml *
---
- name: Déploiement complet sur les VMs Vagrant
  hosts: all
  become: true
  roles:
    - docker_portainer_install
    - jenkins_install
    - container_deployment
```

5. Crée le fichier Vagrantfile

```
 GNU nano 7.2                                                Vagrantfile *
Vagrant.configure("2") do |config|
  # On utilise la même box que pour la démo
  config.vm.box = "bento/ubuntu-24.04"

  # ========================================
  # MACHINE 1 : web1
  # ========================================
  config.vm.define "web1" do |web1|
    web1.vm.hostname = "web1"

    # Réseau privé (permet aux VMs de se voir entre elles)
    web1.vm.network "private_network", ip: "192.168.56.10"

    # Redirection de ports (Hôte de la VM de lab -> VM Vagrant)
    web1.vm.network "forwarded_port", guest: 8082, host: 8082 # Site web
    web1.vm.network "forwarded_port", guest: 8081, host: 8081 # Jenkins
    web1.vm.network "forwarded_port", guest: 9000, host: 9000 # Portainer

    # Ressources allouées à la VM
    web1.vm.provider "virtualbox" do |vb|
      vb.name = "vagrant-web1"
      vb.memory = 2048
      vb.cpus = 2
    end

    # Provisionnement avec Ansible
    web1.vm.provision "ansible" do |ansible|
      ansible.playbook = "playbook.yml"
    end
```

6.Lancement du Lab :

```
arij@vagrant:~/vagrant_lab$ vagrant up
Bringing machine 'web1' up with 'virtualbox' provider...
Bringing machine 'web2' up with 'virtualbox' provider...
==> web1: Importing base box 'bento/ubuntu-24.04'...
==> web1: Matching MAC address for NAT networking...
==> web1: Checking if box 'bento/ubuntu-24.04' version '202510.26.0' is up to date...
==> web1: Setting the name of the VM: vagrant-web1
==> web1: Clearing any previously set network interfaces...
==> web1: Preparing network interfaces based on configuration...
```

```
PLAY [Déploiement complet sur les VMs Vagrant] *********************************

TASK [Gathering Facts] *********************************************************
ok: [web1]

TASK [docker_portainer_install : Installer les prérequis et Docker] ************
changed: [web1]

TASK [docker_portainer_install : Démarrer Docker] ******************************
ok: [web1]

TASK [docker_portainer_install : Ajouter l'utilisateur au groupe docker] *******
changed: [web1]

TASK [docker_portainer_install : Lancer Portainer] *****************************
changed: [web1]

TASK [jenkins_install : Lancer Jenkins (en conteneur Docker)] ******************
changed: [web1]

TASK [container_deployment : Créer le dossier de l'application] ****************
changed: [web1]

TASK [container_deployment : Copier le Dockerfile, l'index.html et le compose] ***
changed: [web1] => (item=Dockerfile)
changed: [web1] => (item=index.html)
changed: [web1] => (item=docker-compose.yml)

TASK [container_deployment : Lancer le conteneur web (build et up)] ************
ok: [web1]

PLAY RECAP *********************************************************************
web1                       : ok=9    changed=6    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
TASK [Gathering Facts] *********************************************************
ok: [web2]

TASK [docker_portainer_install : Installer les prérequis et Docker] ************
changed: [web2]

TASK [docker_portainer_install : Démarrer Docker] ******************************
ok: [web2]

TASK [docker_portainer_install : Ajouter l'utilisateur au groupe docker] *******
changed: [web2]

TASK [docker_portainer_install : Lancer Portainer] *****************************
changed: [web2]

TASK [jenkins_install : Lancer Jenkins (en conteneur Docker)] ******************
changed: [web2]

TASK [container_deployment : Créer le dossier de l'application] ****************
changed: [web2]

TASK [container_deployment : Copier le Dockerfile, l'index.html et le compose] ***
changed: [web2] => (item=Dockerfile)
changed: [web2] => (item=index.html)
changed: [web2] => (item=docker-compose.yml)

TASK [container_deployment : Lancer le conteneur web (build et up)] ************
ok: [web2]

PLAY RECAP *********************************************************************
web2                       : ok=9    changed=6    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

7.Status vagrant :

```
arij@vagrant:~/vagrant_lab$ vagrant status
Current machine states:

web1                      running (virtualbox)
web2                      running (virtualbox)

This environment represents multiple VMs. The VMs are all listed
above with their current state. For more information about a specific
VM, run `vagrant status NAME`.
arij@vagrant:~/vagrant_lab$
```

8.Vérifier les services sur navigateur

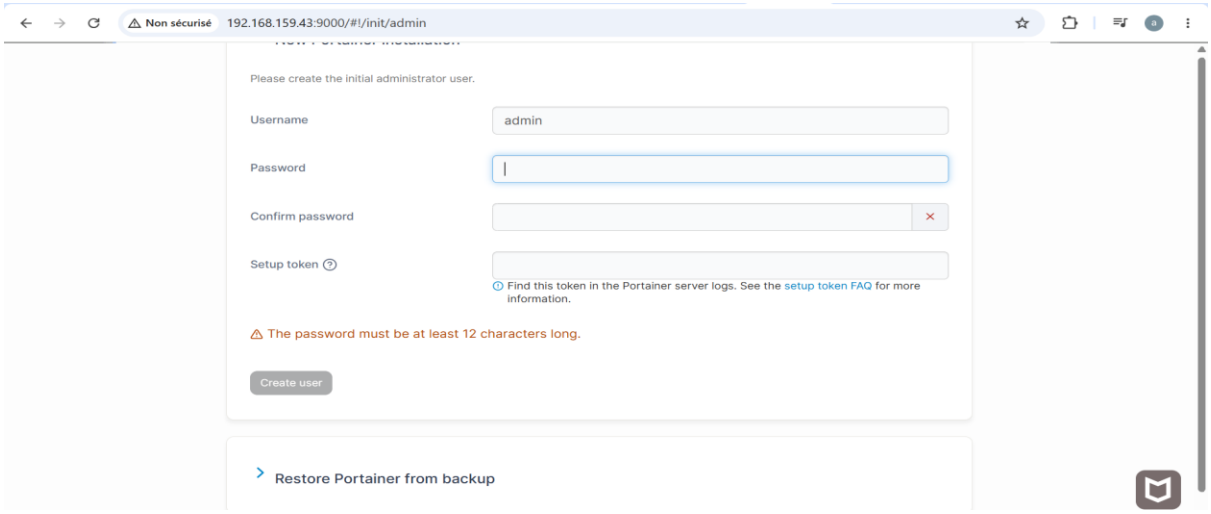
Pour la VM web1

Le site web : http://localhost:8082



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Portainer : http://localhost:9000

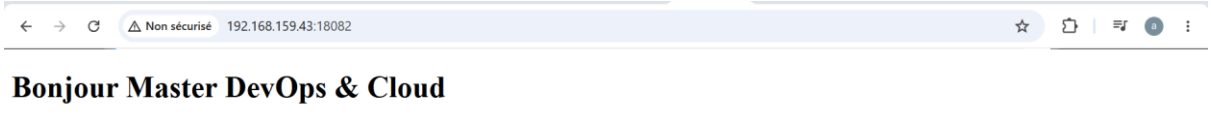


Jenkins : http://localhost:8081

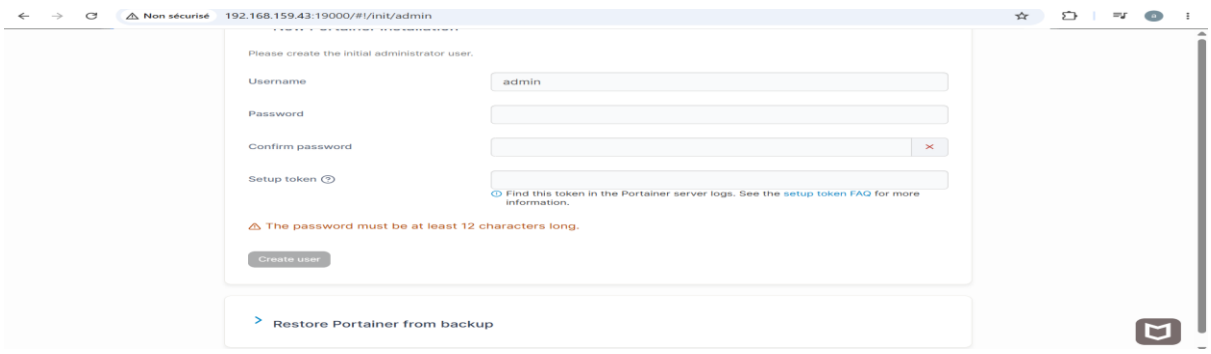


Pour la VM web2 :

Le site web : http://localhost:18082



Portainer : http://localhost:19000



Jenkins : http://localhost:18081

